EXPERIMENT - 2

AIM:- Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

SAMPLE CODE:-

```
-- Create Database
CREATE DATABASE Indian_ECommerce;
USE Indian_ECommerce;
-- 1. CUSTOMER
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Middle_Name VARCHAR(50),
    Last_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Date_of_Birth DATE,
    Address VARCHAR(50) NOT NULL
    );
-- 2. SELLER
CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone_No. VARCHAR(15) NOT NULL UNIQUE,
    Bank_Account VARCHAR(30) UNIQUE
  );
-- 3. CATEGORY
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(255)
  );
```

```
-- 4. ADDRESS
CREATE TABLE Address (
Address_ID INT PRIMARY KEY,
Customer_ID INT NOT NULL,
Address_Line VARCHAR (255) NOT NULL,
City VARCHAR (50) NOT NULL,
Landmark VARCHAR (100),
Address_Type VARCHAR (20),
FOREIGN KEY (Customer_ID)
    REFERENCES
Customer (Customer_ID)
    ON DELETE CASCADE
    ON UPDATE CASCADE
);
-- 5. PRODUCT
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR (100) NOT NULL,
    Price DECIMAL (10,2) NOT NULL,
    Stock_Quantity INT NOT NULL,
    Category_ID INT NOT NULL,
    Seller_ID INT,
    FOREIGN KEY (Category_ID)
        REFERENCES Category (Category_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
  FOREIGN KEY (Seller_ID)
        REFERENCES Seller (Seller_ID)
        ON DELETE SET NULL
        ON UPDATE CASCADE
    );
```

```
-- 6. PRODUCT SPECIALIZATION - ELECTRONICS
CREATE TABLE Electronics (
    Product_ID INT PRIMARY KEY,
    Brand VARCHAR(50) NOT NULL,
    Warranty_Period INT,

    FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- 7. PRODUCT SPECIALIZATION - FASHION
CREATE TABLE Fashion (
    Product_ID INT PRIMARY KEY,
    Size VARCHAR(20),
    Color VARCHAR(30),

    FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- 8. PRODUCT SPECIALIZATION - FOOD
CREATE TABLE Food (
    Product_ID INT PRIMARY KEY,
    Manufacture_DATE DATE,
    Expiry_DATE DATE,

    FOREIGN KEY (PRODUCT_ID)
        REFERENCES Product(Product_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

```
-- 9. ORDERS
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Order_Date DATE NOT NULL,
    Order_Status VARCHAR(30) NOT NULL,
    Total_Amount DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (CUSTOMER_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);

-- 10. ORDER ITEM
-- Composite Primary Key: (Order_ID, Product_ID)
CREATE TABLE OrderItem (
    Order_ID INT,
    Product_ID INT,
    Unit Price DECIMAL(10,2) NOT NULL,
    Quantity INT NOT NULL,
    Discount DECIMAL(5,2) DEFAULT 0,
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE.
        ON UPDATE CASCADE
);

-- 11. PAYMENT
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL,
    Transaction_ID VARCHAR(50) NOT NULL UNIQUE,
    Payment-Method VARCHAR(30) NOT NULL,
    Payment_Status VARCHAR(30) NOT NULL,
    SELLER_ID INT,
```

```
FOREIGN KEY (Order_ID)
    REFERENCES Orders (Order_ID)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

FOREIGN KEY (Seller_ID)
    REFERENCES Seller (Seller_ID)
    ON DELETE SET NULL,
    ON UPDATE CASCADE
);
-- 12. DELIVERY
 CREATE TABLE DELIVERY (
    DELIVERY_ID INT PRIMARY KEY,
    ORDER_ID INT NOT NULL,
    Address_ID INT,

    Delivery_Date DATE,
    Courier_Partner VARCHAR (50),
    Delivery_Status VARCHAR (30),
    Transaction_ID VARCHAR (50),

    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    FOREIGN KEY (Address_ID)
        REFERENCES Address (Address_ID)
        ON DELETE SET NULL
        ON UPDATE CASCADE
    );
```

```
-- INSERT SAMPLE DATA

-- CUSTOMER
INSERT INTO Customer ( Customer_ID, First_Name, Middle_Name,
Last_Name, Email , Date_of_Birth )
VALUES
(101, 'Anushka', NULL, 'Rawat', 'anushkarawat01@gmail.com',
'2005-05-23'),
(102, 'Apurwa', NULL, 'Rawat', 'apurwa15@gmail.com',
'2006-07-15'),
(103, 'Virat', 'Kumar', 'Singh', 'virat16@gmail.com',
'2004-11-15');

-- SELLER
INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone_No,
Bank_Account) VALUES
(201, 'Tech World India', 'techworld@gmail.com',
'9876543210', 'SBIN0012345'),
(202, 'Fashion Hub', 'fashionhub@gmail.com', '9876543211',
'HDFC0005678 9'),
(203, 'Fresh Foods India', 'freshfood@gmail.com',
'9876543212', 'ICIC0009876 5');

-- CATEGORY
INSERT INTO Category ( Category_ID, Category_Name, Description)
VALUES
(301, 'Electronics', 'Electronic devices and accessories'),
(302, 'Fashion', 'Clothing and fashion products'),
(303, 'Food', 'Packaged food products');

-- ADDRESS
INSERT INTO Address ( Address_ID, Customer_ID, Address_Line,
City, Landmark, Address_Type)
VALUES
(401, 101, '12 MG Road', 'Indore', 'Near Metro Station',
```

'Home'),
(402, 102, '45 Civil Lines', 'Lucknow', 'Near Mall', 'Home'),
(403, 103, '22 FC Road', 'Pune', 'Near College', 'Hostel');

-- PRODUCT
INSERT INTO Product (Product_ID, Product_Name, Price, Stock_Quantity, Category_ID, Seller_ID)
VALUES
(501, 'Laptop', 65000.00, 20, 301, 201),
(502, 'T-Shirt', 799.00, 50, 302, 202),
(503, 'Organic Honey', 450.00, 100, 303, 203);

-- ELECTRONICS
INSERT INTO Electronics (Product_ID, Brand, Warranty_Period)
VALUES (501, 'HP', 24);

-- FASHION
INSERT INTO Fashion (Product_ID, Size, Color) VALUES (502, 'S', 'Blue');

-- FOOD
INSERT INTO Food (Product_ID, Manufacture_Date, Expiry_Date)
VALUES
(503, '2026-07-01', '2027-07-01');

--ORDERS
INSERT INTO Orders (Order_ID, Customer_ID, Order_Date, Order_Status, Total_Amount)
VALUES
(601, 101, '2026-08-15', 'Confirmed', 65799.00),
(602, 102, '2026-08-16', 'Shipped', 799.00);

-- ORDER ITEMS
INSERT INTO OrderItem (Order_ID, Product_ID, Unit_Price, Quantity, Discount)
VALUES
(601, 501, 65000.00, 1, 0.00),
(601, 502, 799.00, 1, 0.00),
(602, 502, 799.00, 1, 50.00);

-- PAYMENT
INSERT INTO Payment (Payment_ID, Order_ID, Transaction_ID, Payment_Method, Payment_Status, Seller_ID)

```
VALUES
(701, 601, 'TXN10001', 'UPI', 'Successful', 201),
(702, 602, 'TXN10002', 'Card', 'Successful', 202);

-- DELIVERY
INSERT INTO Delivery (Delivery_ID, Order_ID, Address_ID, Delivery_Date,
Courier_Partner, Delivery_Status, Transaction_ID)
VALUES
(801, 601, 401, '2026-08-20', 'BlueDart', 'Out for Delivery',
'TXN10001'),
(802, 602, 402, '2026-08-21', 'BlueDart', 'Shipped',
'TXN10002');
-- TEST 1: REFERENTIAL INTEGRITY VIOLATION
-- Invalid Customer_ID = 999
-- This statement should give a FOREIGN KEY Error.
INSERT INTO Orders (Order_ID, Customer_ID, Order_Date,
Order_Status, Total_Amount) VALUES
(603, 999, '2026-08-20', 'Confirmed', 5000.00);
-- TEST 2: REFERENTIAL INTEGRITY VIOLATION
-- Invalid Product_ID = 999
-- This statement should give a FOREIGN KEY error.
INSERT INTO OrderItem (Order_ID, Product_ID, Unit_Price, Quantity,
Discount) VALUES (601, 999, 500.00, 1, 0.00)
-- DISPLAY TABLES
SELECT * FROM Customer;
SELECT * FROM Seller;
SELECT * FROM Category;
SELECT * FROM Address;
SELECT * FROM Product;
SELECT * FROM Electronics;
SELECT * FROM Fashion;
SELECT * FROM Food;
SELECT * FROM Orders;
SELECT * FROM OrderItem;
SELECT * FROM Payment;
SELECT * FROM Delivery;
```

OUTPUT :-

CUSTOMER

Customer_id	First_Name	Middle_Name	Last_Name	Emailid	Date_of_Birth
101	Anushka	Null	Rawat	anushkarawat01@gmail.com	2005-05-23
102	Apurwa	Null	Rawat	apurwa15@gmail.com	2006-07-15
103	Virat	Kumar	Singh	virat16@gmail.com	2004-11-15

